

SherpAI V1 Whitepaper

Terrain Intelligence Operator for TerraSatch

Version v1.0.0 — March 05, 2026



Prepared for

Avalanche & mountain safety organizations • Land managers • Infrastructure & utility operators • Ski areas • SAR teams • Municipalities • Insurers • Outdoor brands • Technical partners

0. Front Matter

0.1 Title Page (version, date)

Document Title: SherpAI V1 Whitepaper: Terrain Intelligence Operator for TerraSatch

Version: v1.0.0

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0.2 Disclaimer (safety/legal)

TerraSatch and SherpAI provide decision-support software intended to improve terrain awareness and operational coordination. They do not prevent accidents and do not replace professional training, certified avalanche education, local advisories, incident command, or organizational safety protocols. Avalanche terrain, weather, snowpack, wildfire behavior, hydrology, slope stability, and infrastructure exposure involve uncertainty, incomplete information, and rapidly changing conditions. Users must verify outputs against field observations, official forecasts and warnings, and site-specific expertise. SherpAI is not a generic chatbot. It is a domain-constrained operator layer that produces evidence-linked briefings and map overlays. SherpAI does not issue deterministic safe/unsafe or go/no-go decisions and will refuse requests that imply such guarantees. This whitepaper describes V1 scope and roadmap. Unless explicitly stated as currently available in V1, features described as planned or future are V1 proposals rather than commitments.

0.3 Contact / Links

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Contact: hello@terrasatch.com

Docs: <https://www.terrasatch.com/docs>

Pricing: <https://www.terrasatch.com/pricing>

Table of Contents

0. Front Matter	2
0.1 Title Page (version, date)	2
0.2 Disclaimer (safety/legal)	2
0.3 Contact / Links	2
1. Executive Summary	5
1.1 What SherpAI enables inside TerraSatch	5
1.2 V1 scope boundaries (in scope, next, out of scope)	5
2. Automation Behind the Scenes	5
2.1 The SherpAI Run protocol	5
2.2 AvyTS operational cadence	6
2.3 Alerting and notifications pipeline	6
2.4 Drone/LiDAR evidence to operational products	6
3. Problem Statement	8
3.1 The gap in current workflows	8
3.2 Operational consequences	8
4. Technology Architecture (V1)	8
4.1 Architecture components	8
4.2 Clean schema summaries (operator-relevant)	8
Observation (summary schema)	8
SherpAI Run (summary schema)	9
5. Metrics, Validation, and Operational KPIs	9
5.1 What SherpAI measures (V1)	9
5.2 Example dashboard panels (pilot reporting examples)	9
5.3 Validation approach (conservative, ops-focused)	10
6. Use Cases (SherpAI × TerraSatch Modules)	11
6.1 Backcountry planning and education (AvyTS)	11
6.2 Patrol and resort operations (AvyTS + InfraTS)	11
6.3 Corridor risk and inspections (GeoTS + InfraTS)	11
6.4 Melt/runoff hazards (AvyTS ↔ HydroTS)	11
6.5 Wildfire operational context (PyroTS + InfraTS)	11
7. Roadmap	11
7.1 Phase 0 (now): baseline and pilots	11
7.2 Phase 1 (V1): operational release	11

7.3 Phase 2 (V2): deeper coupling and comms	11
7.4 Phase 3 (scale): multi-region operations	12
8. Safety, Limits, and Governance	12
8.1 What SherpAI will not do	12
8.2 Governance controls	12
9. Appendix	12
9.1 References	12
9.2 Example visual: AvyTS risk-scale briefing surface	12

1. Executive Summary

SherpAI is TerraSatch's terrain intelligence operator: a domain-constrained, provenance-aware, safety-gated decision-support layer that orchestrates TerraSatch modules (AvyTS, GeoTS, HydroTS, PyroTS, InfraTS) into operational map overlays, briefs, alerts, and repeatable run history.

V1 is intentionally conservative. SherpAI prioritizes evidence linkage, uncertainty-first communication, and workflows that align with professional protocols. SherpAI integrates authoritative advisories and partner policy gates rather than replacing them.

1.1 What SherpAI enables inside TerraSatch

- **Unified operational picture:** converts fragmented inputs into a time-specific, location-specific briefing.
- **Evidence-linked outputs:** every overlay or brief lists inputs used, quality flags, and deltas since the last run.
- **Run history:** reproducible snapshots (Run IDs) for after-action review, training, and governance.
- **Cross-module reasoning:** connects AvyTS ↔ HydroTS, GeoTS ↔ InfraTS, and PyroTS ↔ InfraTS where appropriate.
- **Safety gates:** refusal rules for deterministic claims plus configurable organization policy gates for publishing and sharing.

1.2 V1 scope boundaries (in scope, next, out of scope)

In scope (V1): Runs and Evidence Packs; provenance scoring; module interpreters and conservative fusion; gradients and overlays; corridor and route summaries; observation prompting; review gates; share links tied to snapshots; exports and APIs.

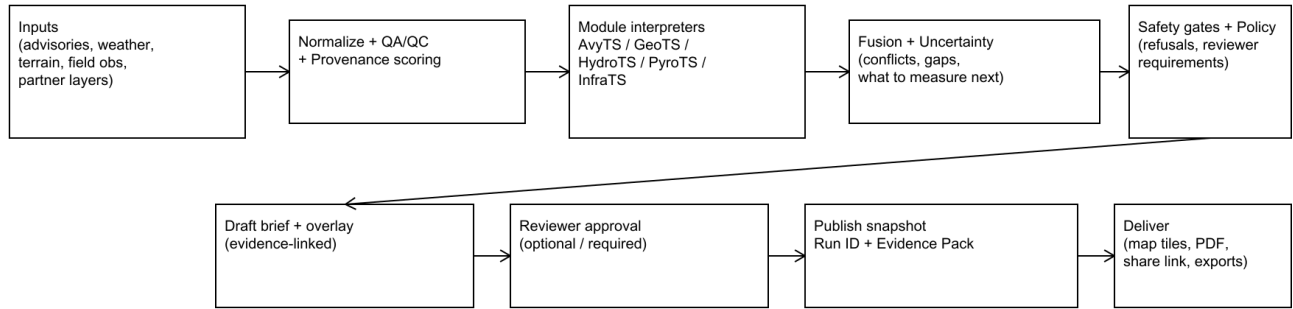
Next (post-V1): deeper AvyTS–HydroTS coupling; expanded GeoTS corridor pilots; structured comms logging (radio/voice → observations); calibration dashboards; partner webhooks where contracted.

Out of scope (V1): automated go/no-go decisions; guarantees; black-box models without audit trails; universal calibration without partner validation; overriding official warnings.

2. Automation Behind the Scenes

2.1 The SherpAI Run protocol

SherpAI automation is expressed as a repeatable Run protocol. Runs standardize inputs, enforce provenance and QA/QC, apply module interpreters, and generate evidence-linked products with explicit uncertainty and safety gates.

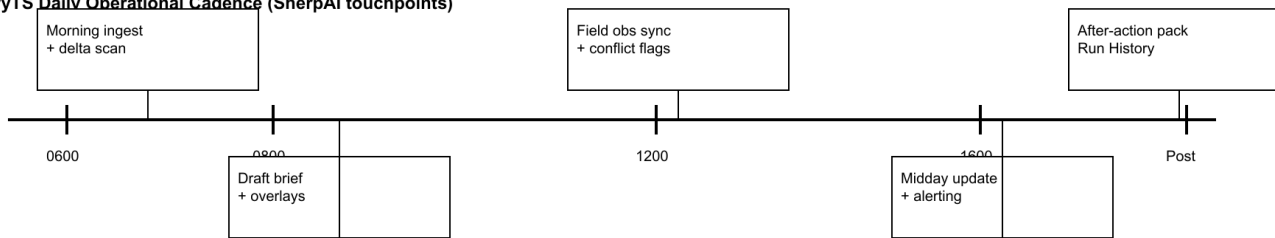


Every published product is tied to a Run ID and Evidence Pack so decisions can be reviewed and reproduced.

2.2 AvyTS operational cadence

For AvyTS, SherpAI is designed to match the cadence of daily avalanche decision-support: morning synthesis, field updates, mid-day deltas, and after-action packaging.

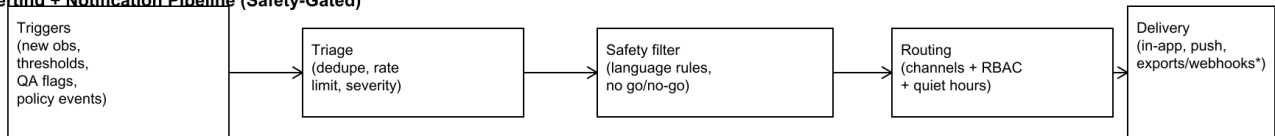
AvyTS Daily Operational Cadence (SherpAI touchpoints)



2.3 Alerting and notifications pipeline

SherpAI supports safety-aware alerts designed to reduce noise and prevent overconfident language. Alerts are evidence-linked and governed by RBAC and organization policy.

Alerting + Notification Pipeline (Safety-Gated)

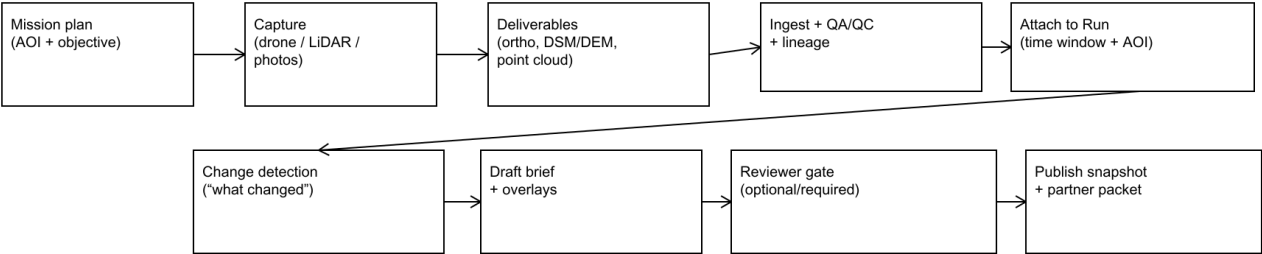


**Webhooks/exports enabled by plan and partner agreements.*

2.4 Drone/LiDAR evidence to operational products

When drone or LiDAR products are available (public or partner-provided), SherpAI treats them as evidence layers with lineage, QA/QC, and explicit applicability notes—then translates them into Run deltas and decision-support products.

Drone / LiDAR Capture → SherpAI Evidence → Operational Products



3. Problem Statement

3.1 The gap in current workflows

Teams can access advisories, terrain layers, weather products, and field notes—but the missing piece is an operator workflow that standardizes inputs, forces provenance, communicates uncertainty, and produces repeatable briefs with defensible run history.

3.2 Operational consequences

- Conditions shift faster than teams can reconcile across tools.
- Briefs and incident packages are difficult to recreate after the fact.
- Without deltas and evidence linkage, teams over-weight stale data or miss key changes.

4. Technology Architecture (V1)

4.1 Architecture components

- **Ingestion and normalization:** geospatial layers, advisories, weather feeds, telemetry where available, and partner layers.
- **Terrain foundation (TerraGrid):** consistent derivatives used across modules.
- **Module interpreters:** hazard-logic pipelines (AvyTS/GeoTS/HydroTS/PyroTS/InfraTS).
- **Fusion and uncertainty:** conflict detection and conservative aggregation.
- **Publishing and governance:** draft/publish workflows, reviewer approvals, permissioned sharing, and audit logs.
- **Exports and APIs:** GIS exports and integration surfaces for partners.

4.2 Clean schema summaries (operator-relevant)

Observation (summary schema)

Field	Description	Notes
observation_id	Unique identifier for the observation.	Stable across sync.
created_at	Timestamp in UTC.	Used for run windows.
location	Point geometry (lon/lat) plus optional elevation.	Offline-capable.
type	Category (avalanche_snowpack, rockfall, wildfire, infrastructure, etc.).	Controls fields.
tags	Standardized tags and hazard markers.	Filter/search.
summary	One-sentence narrative.	Human-readable.
details	Minimal structured fields by type.	V1 conservative.
media	Linked evidence (photo/audio/video).	Evidence Pack.

review_status	draft/pending/published plus notes.	Policy gate.
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SherpAI Run (summary schema)

Field	Description	Notes
run_id	Unique run identifier.	Appears in briefs.
created_at	Run timestamp.	Audit.
aoi	Zone/corridor/route reference.	Scope.
time_window	Start/end timestamps.	Input selection.
inputs_used	Inputs plus provenance flags.	Evidence anchor.
modules	Interpreters executed.	AvyTS-first in V1.
uncertainty	Low/medium/high plus notes.	Displayed.
outputs	Overlay/brief/snapshot IDs.	Reproducible.
review	Draft/published plus reviewer requirements.	Governance.

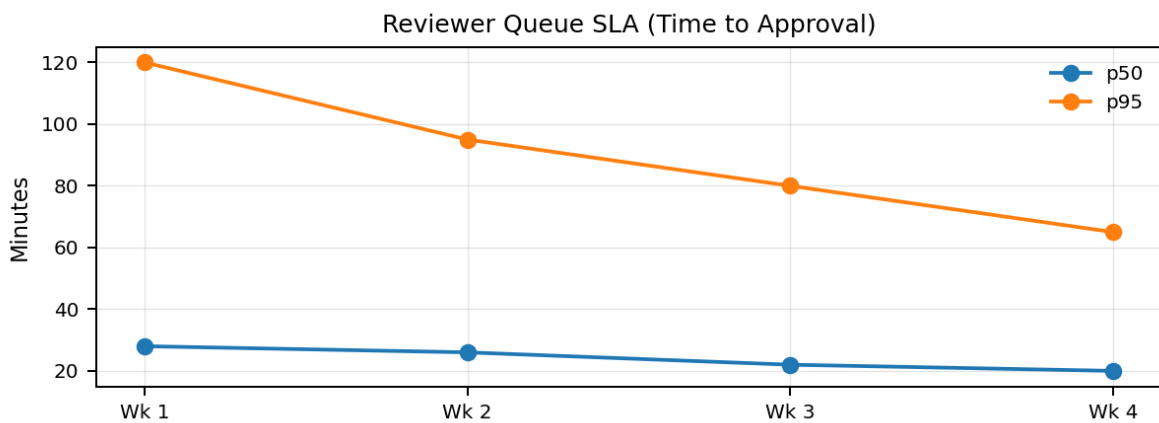
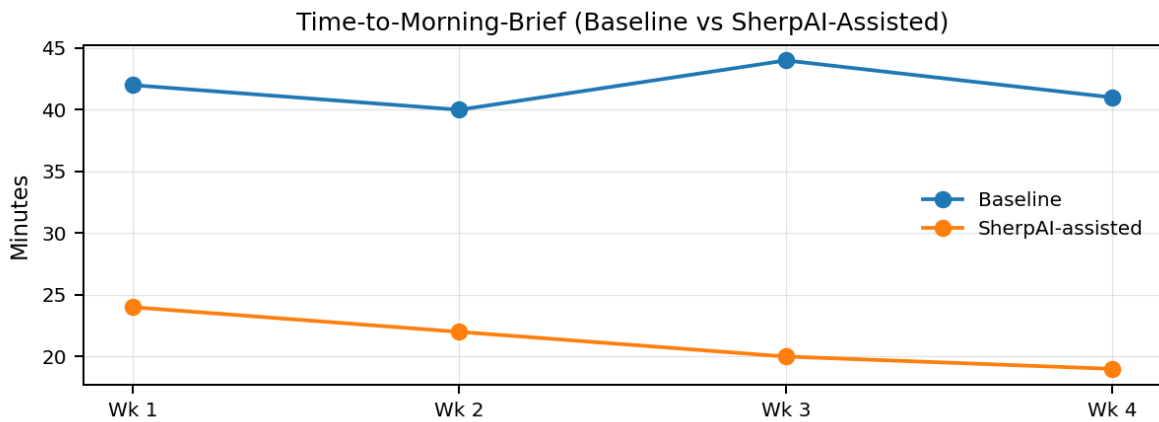
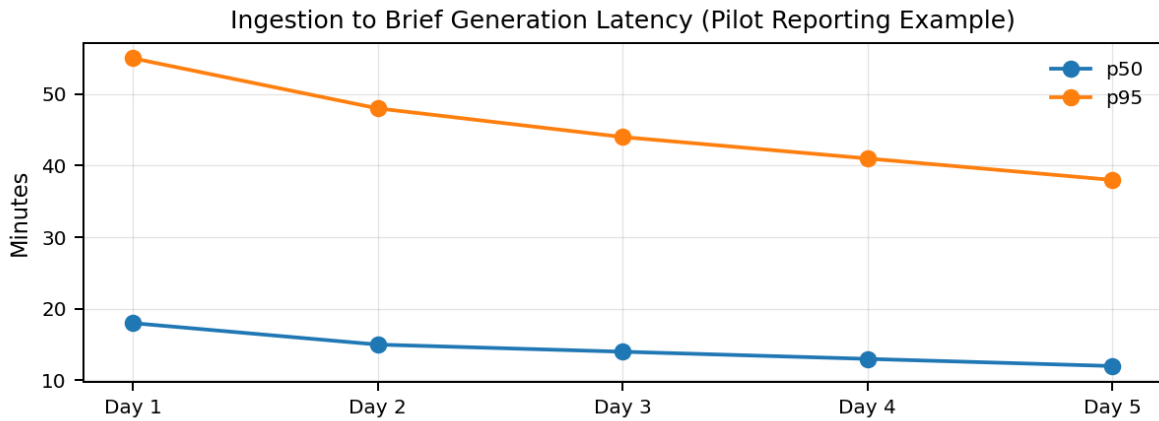
5. Metrics, Validation, and Operational KPIs

5.1 What SherpAI measures (V1)

- **Reliability and timeliness:** ingestion latency, overlay availability, sync success rates, alert latency.
- **Workflow performance:** time-to-morning-brief, reviewer queue SLA, percentage of outputs published with complete Evidence Packs.
- **Decision-support health:** conflict rate, data coverage scores, reviewer-labeled false alarm rate where labels exist.

5.2 Example dashboard panels (pilot reporting examples)

The charts below are presented as pilot reporting examples to illustrate how SherpAI operational performance is measured. Metrics are organization-specific and depend on data availability and governance requirements.



5.3 Validation approach (conservative, ops-focused)

- **Backtesting where labels exist:** compare prior-season Runs to archived advisories and observed outcomes, with explicit limits on label quality.
- **Expert review workflow:** forecaster and operator review gates for high-impact overlays; feedback tied to Run IDs.
- **Field trials:** measure time saved, briefing consistency, and trust outcomes; avoid claims of incident prevention.
- **Model change control:** versioned rollouts with evaluation reports and rollback capability.

6. Use Cases (SherpAI × TerraSatch Modules)

6.1 Backcountry planning and education (AvyTS)

- Route corridor summaries by elevation/aspect bands with evidence-linked callouts.
- Observation prompts to reduce uncertainty (transport signs, surface warming, stability test results).
- Education mode: conservative, non-alarmist language aligned with standard hazard terminology.

6.2 Patrol and resort operations (AvyTS + InfraTS)

- Daily briefs aligned to shift cadence with deltas since the last run.
- Reviewer gates for publishing; permissioned share links for teams.
- Incident packages tied to exact snapshots for reproducibility.

6.3 Corridor risk and inspections (GeoTS + InfraTS)

- Screening-level rockfall/slope overlays to prioritize inspections.
- Asset segment exposure views and evidence packs for planning.
- Drone/LiDAR ingestion where available under partner agreements.

6.4 Melt/runoff hazards (AvyTS ↔ HydroTS)

- Wet instability timing windows and runoff-driven exposure planning.
- Crossing/corridor watch zones; telemetry-aware constraints when data is sparse.

6.5 Wildfire operational context (PyroTS + InfraTS)

- Wind-shift exposure planning with time-window watch briefs.
- Asset/corridor prioritization with conservative uncertainty communication.

7. Roadmap

7.1 Phase 0 (now): baseline and pilots

- Brief Builder and share links tied to reproducible snapshots.
- Offline-first observation capture and sync.
- AvyTS-first interpreter and conservative overlays.
- Run History, Evidence Packs, and audit log baseline.

7.2 Phase 1 (V1): operational release

- Standardized Runs for zones/corridors/routes with consistent uncertainty output.
- Reviewer workflows for high-impact publishing.
- Export bundles and documented APIs for partners.
- Backtesting baselines and evaluation reports tied to versions.

7.3 Phase 2 (V2): deeper coupling and comms

- AvyTS–HydroTS coupling for melt/runoff and wet instability signals.

- Expanded GeoTS corridor pilots with partner evidence inventories.
- Structured comms logging (radio/voice → timestamped observations).
- Calibration dashboards and safe rollout/rollback tooling.

7.4 Phase 3 (scale): multi-region operations

- Repeatable ingestion playbooks and onboarding kits for new regions/partners.
- Reliability targets for tiles, sync success, and alert latency.
- Integration templates and webhooks where contracted.

8. Safety, Limits, and Governance

8.1 What SherpAI will not do

- No deterministic “safe/unsafe” declarations.
- No authoritative go/no-go instructions.
- No overriding official warnings and closures.
- No claims of incident prevention or guarantees.

8.2 Governance controls

- Draft vs published layers with reviewer sign-off for high-severity outputs.
- Permissioned sharing with audit logs (who shared which snapshot and when).
- Run History retained for after-action review and training.

9. Appendix

9.1 References

TerraSatch overview and module descriptions: <https://terrasatch.com>. Documentation: <https://www.terrasatch.com/docs>. Pricing: <https://www.terrasatch.com/pricing>.

9.2 Example visual: AvyTS risk-scale briefing surface



SherpAI uses consistent hazard communication and links it to evidence and uncertainty, avoiding simplistic safe/unsafe conclusions.